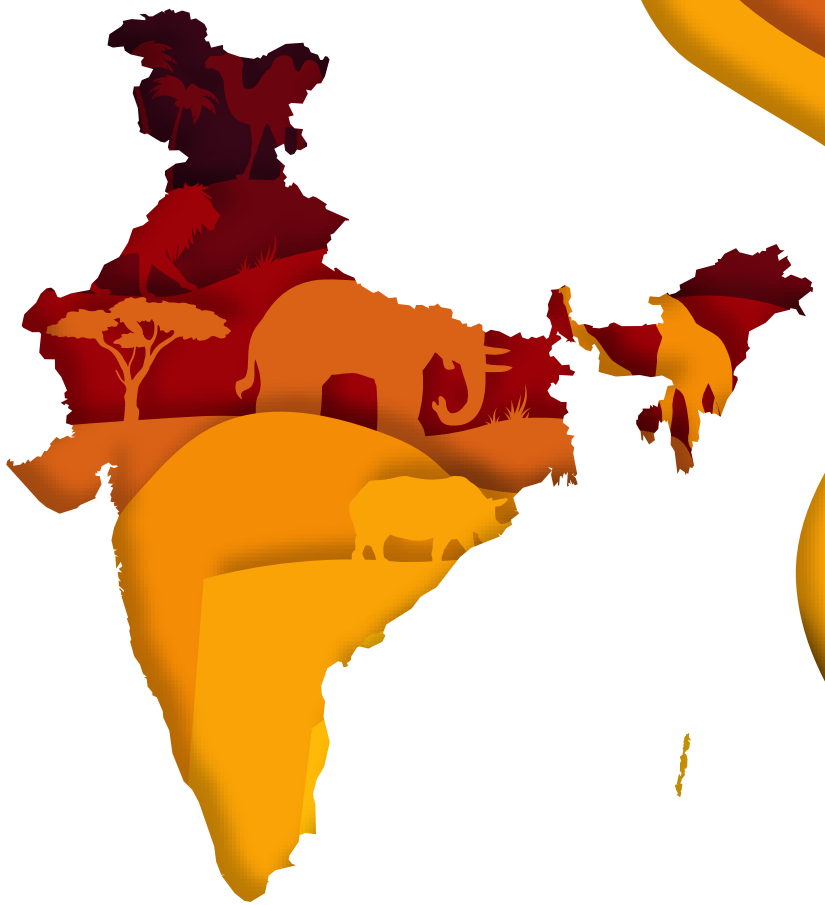


COMPOUND HEAT–DROUGHT STRESS OVER INDIA

Are ecosystems hit harder when heat and drought strike together?

Mid-Evaluation, EVS Project

Arnav Deshmukh · Ayush Brajesh Roy · Saksham Chitkara
Gracy Garg · Rudra Choudhary
IIIT Hyderabad · October 9, 2026

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OUTLINE



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05 THINGS LEFT TO DO

01

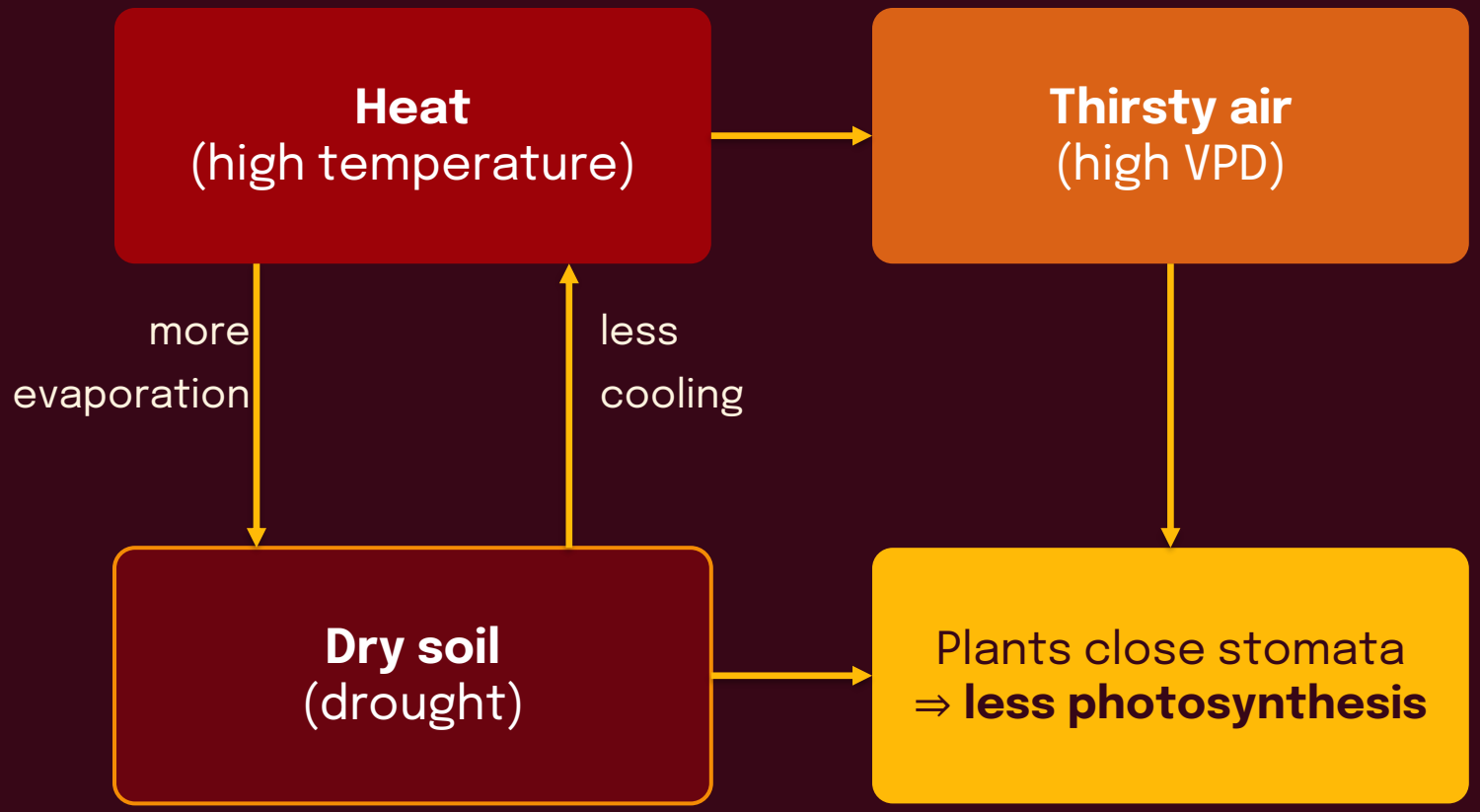
PROBLEM STATEMENT & MOTIVATION



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HEAT AND DROUGHT RARELY COME ALONE

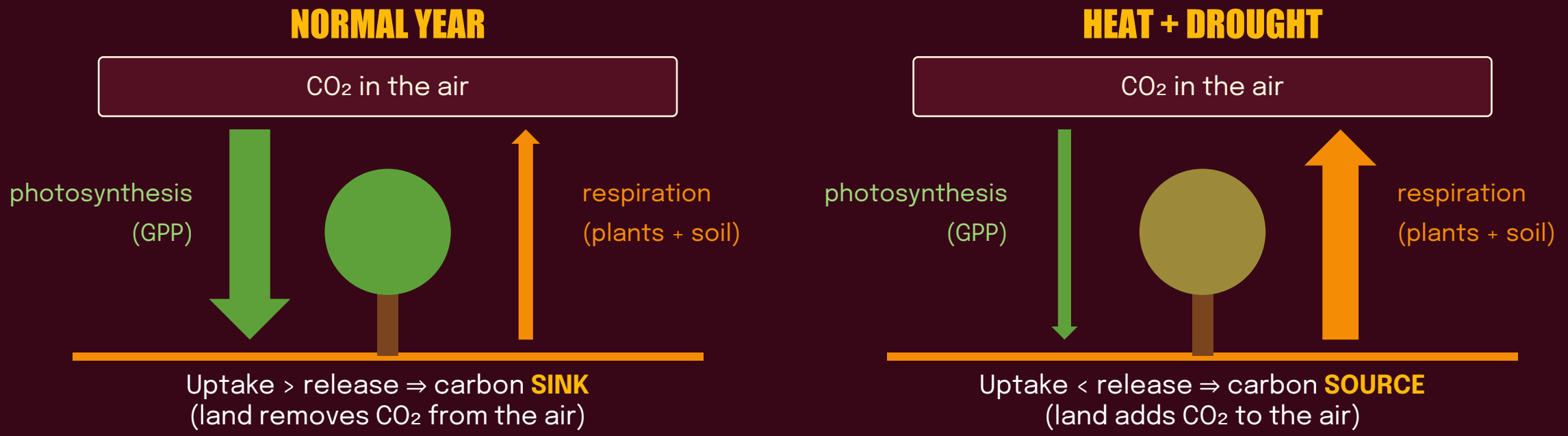
- **Drought**: too little water (little rain, dry soil)
- **Heat**: high temperature, “thirsty” air
- They feed each other: dry soil heats up, heat dries soil
- Plants get hit from both sides



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KEY IDEA: GPP AND THE LAND CARBON SINK

Gross primary productivity (GPP) is the total amount of carbon that plants take out of the air by photosynthesis, per area and time (we use grams of carbon per m^2 per month). It is the starting point of plant growth, crop yield and the land carbon sink.



Heat + drought can flip land from sink to source: Europe in 2003 [1].

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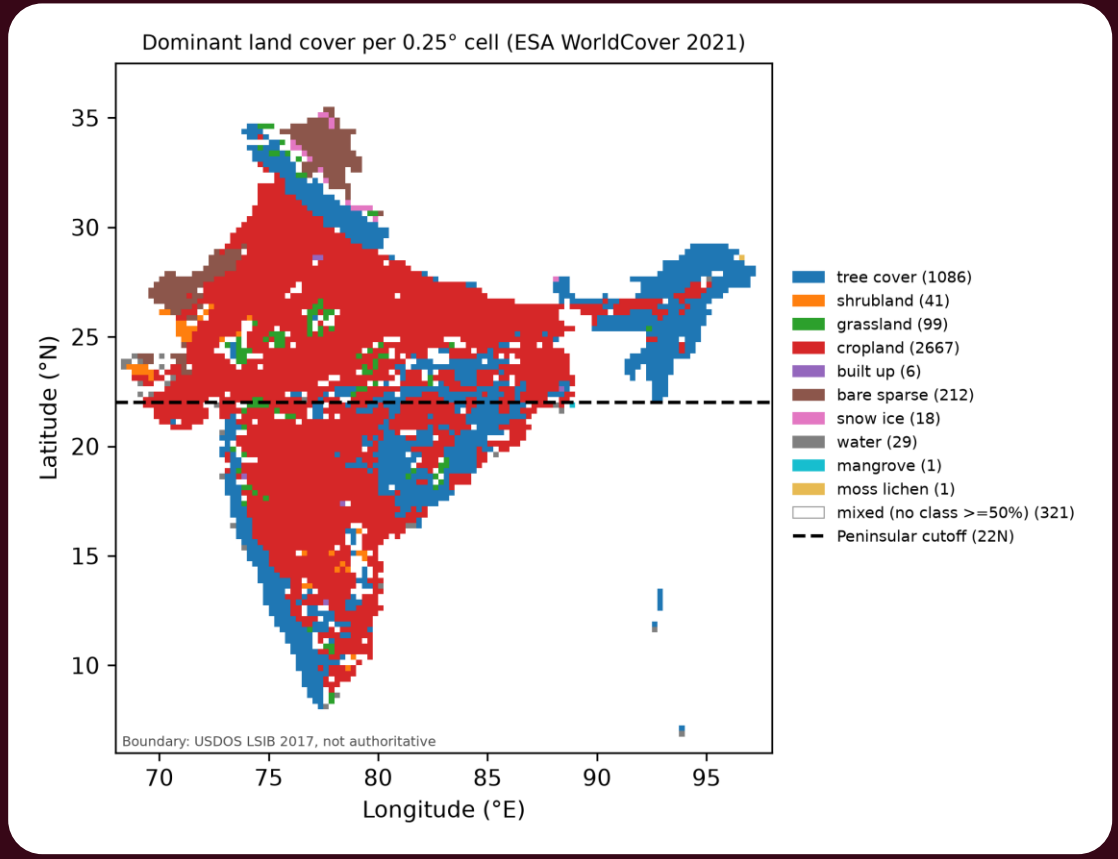
WHY IT MATTERS

- **Europe 2003**: a summer of heat + drought cut productivity by about 30% and turned Europe's land into a temporary carbon *source* [1].
- **Worldwide**: across flux-tower sites, combined heat + drought reduced the carbon sink more than either extreme alone [2].
- **India**: droughts and heatwaves that happen at the same time have become more frequent and widespread [3].
- As the climate warms, more droughts will happen during hot spells, so we need to know how much worse the combination is.

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WHY INDIA?

- **Monsoon-dependent**: most rain falls in Jun–Sep; a weak monsoon means drought (2002, 2009).
- **Hot pre-monsoon**: Mar–May is the hottest season; extreme heat in 2010 and 2022.
- **Mostly cropland** (~60% of cells): lost productivity means lost food and income.
- **Under-studied**: most work covers Europe, China or the globe; India-wide studies are new [4, 5].



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PROBLEM STATEMENT

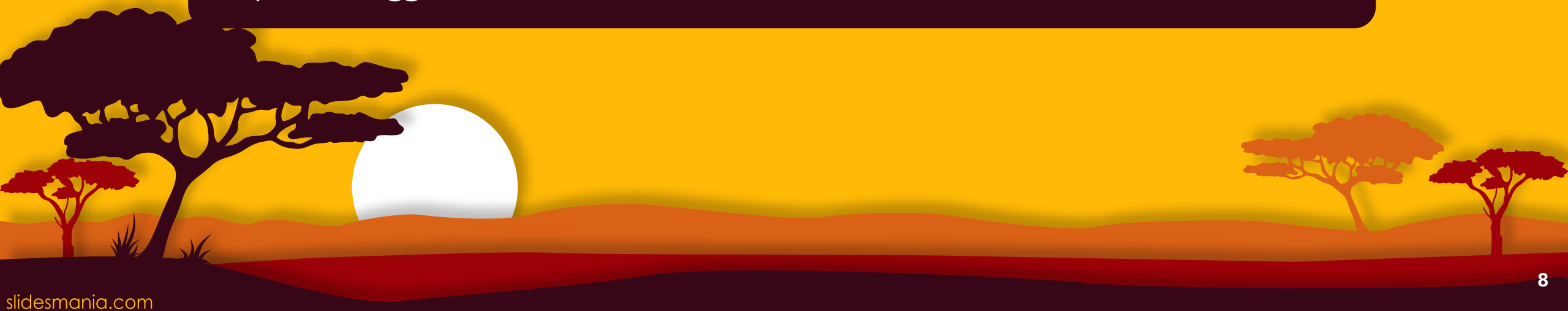
TOPIC: COMPOUND HEAT–DROUGHT STRESS

Are ecosystems experiencing stronger combined impacts?

Does simultaneous heat and drought produce **greater ecosystem stress** than either disturbance alone?

WHY THIS MATTERS

If heat and drought damage simply **add up**, we can predict compound impacts from single-stress studies. If they **amplify** each other, compound events are a separate, bigger risk.



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OUR QUESTIONS (FOR INDIA)

1. Is vegetation hit harder in **compound** (hot *and* dry) months than in hot-only or dry-only months?
2. Is the compound hit **bigger than the heat damage plus the drought damage added together**?
3. Do ecosystems take **longer to recover** after compound events?
4. Does the answer differ between **forests, croplands, grasslands**, and between **Peninsular and North India**?



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WHAT IS KNOWN, AND WHAT IS STILL OPEN

KNOWN

- Heat + drought together cut the carbon uptake of ecosystems more than either extreme alone (flux towers worldwide [2]; Europe 2003 [1]).
- In India, hot and dry conditions together raise the *chance* that vegetation goes into drought [4].
- In Peninsular India's forests, compound hot-dry extremes raise the risk of large GPP losses [5].

OPEN

- Is the damage to *average* productivity more than heat damage + drought damage (synergy), or just the sum? Few studies test this directly, with uncertainty.
- The answer depends on how "drought" is defined: soil-based and air-based definitions give different impacts [6].
- Do ecosystems recover more slowly after compound events? Global studies disagree [7, 8, 9], and we found no India-specific comparison.

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02

OUR HYPOTHESES



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OUR HYPOTHESES

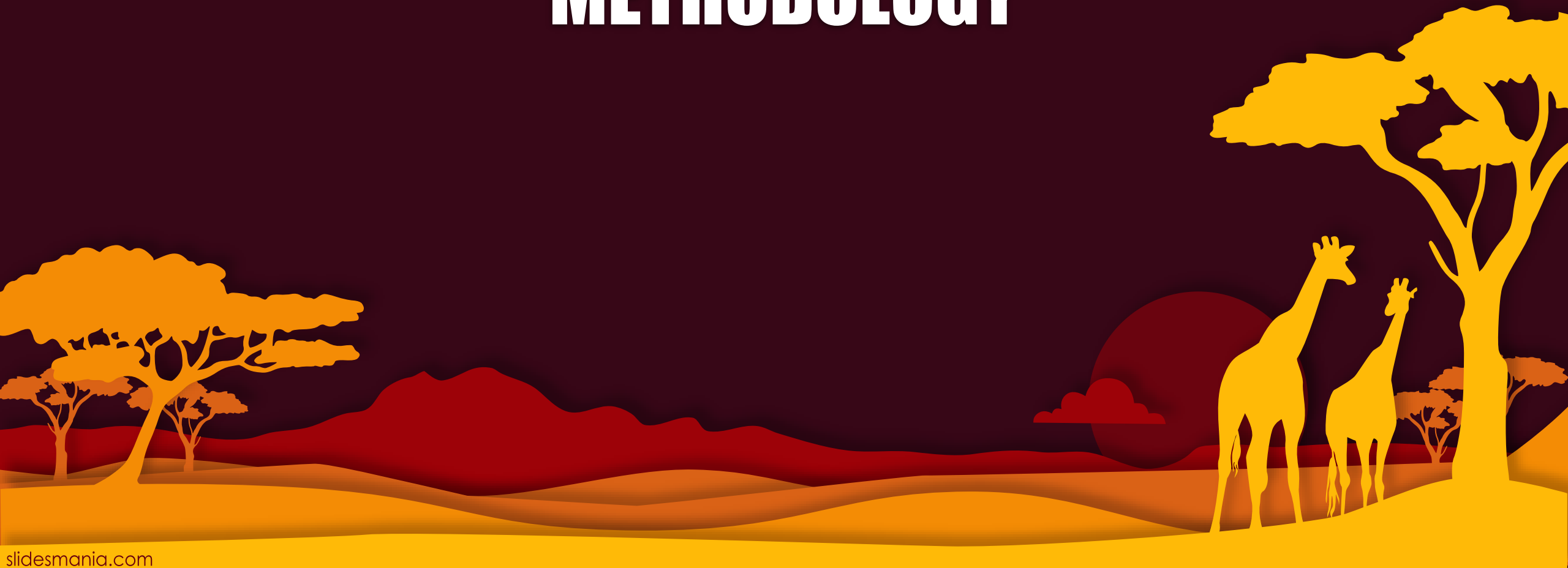
Our aim: an India-wide, honest test of additivity on two satellite measures, with uncertainty and a drought-index check, then recovery and regional differences. This is a regional synthesis with extensions, not a first-of-its-kind study.

	Hypothesis	Current status
H1	Vegetation (GPP, NDVI) is lower in compound months than in hot-only or dry-only months.	**Tested**
H2	The compound effect is *synergistic*: larger than the heat effect + the drought effect.	**Tested**
H3	Ecosystems recover more slowly after compound events than after droughts.	In progress
H4	The response differs by land cover and by region (Peninsular vs North India).	In progress

Throughout, we check that conclusions do not depend on the choice of drought index (SPEI-3 vs SPI-3).

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03
METHODOLOGY



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WORKFLOW

STEP 1
Download & aggregate data (Earth Engine)

→

STEP 2
Monthly anomalies (z-scores)

→

STEP 3
Sort each grid-month: normal / hot / dry / compound

→

STEP 4
Compare GPP & NDVI across boxes (H1, H2)

↓

Events & recovery (H3), land cover & region (H4) *in progress*

Everything is scripted in Python (xarray, scipy, statsmodels) and checked by 68 automated tests. The analysis reruns from cached data in minutes.

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STEP 1: DATA

Dataset	What we use	Period	Role
ERA5-Land [10]	daily max temperature; soil moisture 0–100 cm; VPD	1991–2024	defines “hot”; mechanism
SPEI v2.11 [11]	SPEI-3 (rain minus evaporative demand)	1991–2024	defines “dry”
CHIRPS v2 [12]	rainfall → SPI-3	1991–2024	temperature-free drought check
MODIS MOD17A2H [13, 14]	GPP (photosynthesis)	2001–2022	main response
MODIS MOD13A3 [15]	NDVI (greenness), quality-masked	2001–2022	second response
ESA WorldCover [16]	land-cover fractions	2021	vegetation mask; land-cover groups

- All five suggested datasets, processed in Google Earth Engine [17] onto one common grid: 0.25° (≈27 km), monthly, all of India (4,481 grid cells).
- GPP and NDVI both come from the MODIS Terra satellite: two different products, but not independent sensors.

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METRICS AT A GLANCE

Metric	What it measures	How to read it
Ecosystem response		
GPP	Carbon taken up by plants through photosynthesis (gC/m^2/month)	higher = more productive
NDVI	Canopy greenness from red and near-infrared reflectance (0–1)	higher = greener
Stress		
Tmax z	Anomaly of the monthly mean daily maximum temperature	≥ 1: “hot” month
SPEI-3	3-month water balance: rain minus evaporative demand	≤ -1: “dry” month
SPI-3	3-month rainfall only (no temperature in it)	≤ -1: dry (robustness)
Soil moisture	Water in the top 1 m of soil	lower = drier soil
VPD	Vapour-pressure deficit: how “thirsty” the air is	higher = more water stress
Statistics		
z-score	Distance from normal for that place and calendar month (in SDs)	0 = normal; -1 = clearly low
Amplification	Compound effect − (heat effect + drought effect)	< 0 synergy; ≈ 0 additive
95% CI	Year-block bootstrap: 1,000 resamples of whole years	excludes 0 = significant

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STEP 2: MONTHLY ANOMALIES (COMPARING LIKE WITH LIKE)

- Raw values are not comparable: May is always hotter than January, and a desert always has less GPP than a forest.
- So for every grid cell and every calendar month, we convert a value into a **z-score**:

$$z = \frac{\text{this month's value} - \text{normal for this cell in this month}}{\text{typical year-to-year spread}}$$

  - $z = -1$ means "one standard deviation below what is normal here at this time of year".
- **Temperature**: normal = 1991–2020. Not detrended, so long-term warming counts as heat.
- **GPP, NDVI**: a linear trend is removed first, so the long-term greening of India (CO_2, irrigation) is not mistaken for an anomaly.
- **Only growing-season months count**: the cell is in India, at least 50% vegetated, and GPP in that month is normally at least 30% of its peak. This removes deserts and dormant seasons (824,598 grid-months remain).

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STEP 3: SORTING EVERY GRID-MONTH INTO FOUR BOXES

Tmax z	SPEI-3 (dryness) > −1	SPEI-3 (dryness) ≤ −1
< 1	Normal	Dry-only
≥ 1	Hot-only	**Compound**

Normal	573,543
Hot-only	79,717
Dry-only	109,073
Compound	**62,265** (7.6%)

- **Hot**: the month's average daily maximum temperature is at least 1 standard deviation above normal.
- **Dry**: **SPEI-3** ≤ −1, i.e. the last 3 months had clearly less water than usual (rain minus evaporative demand).
- **SPEI**'s demand term depends on temperature, so "dry" is partly driven by heat. We therefore repeat everything with **SPI-3**, which uses rainfall only.
- Left: number of growing-season grid-months in each box, India, 2001–2022.

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STEP 4: HOW WE TEST H1 AND H2

- **H1**: average GPP *z* (and NDVI *z*) in each box, and the differences between boxes.
- **H2**: compare the compound effect with the sum of the single effects. Each effect is measured relative to normal months:

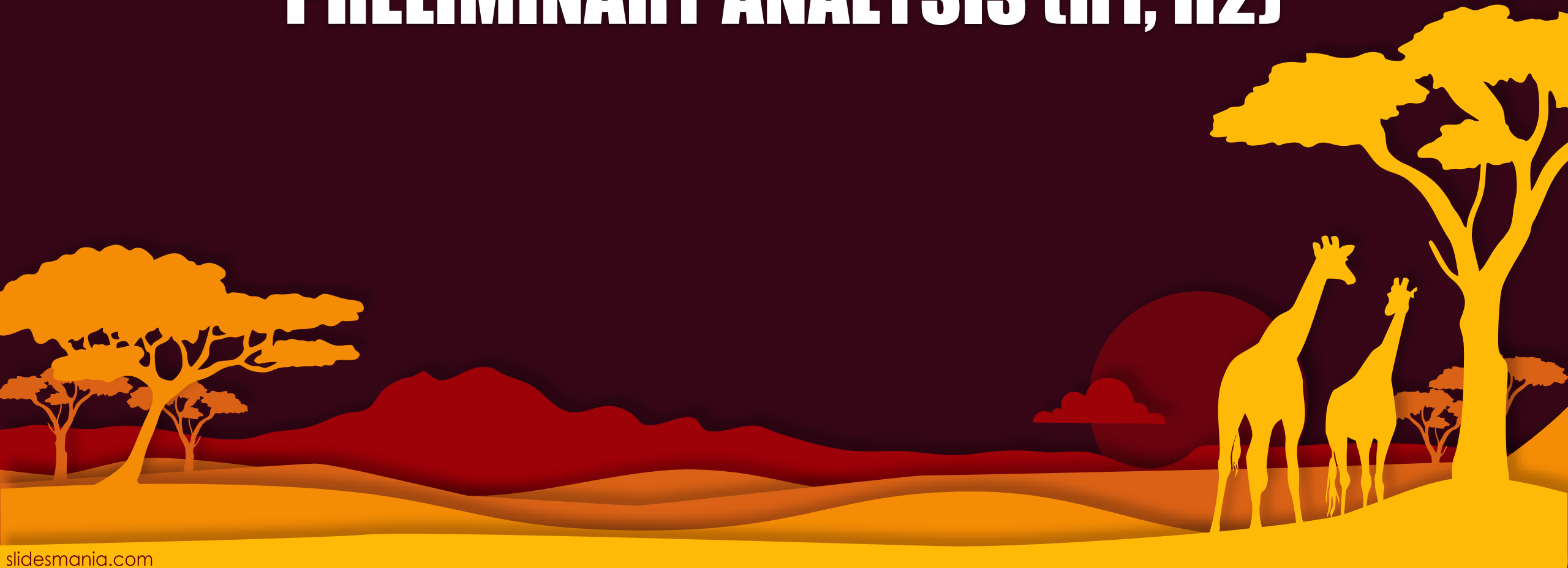
$$\text{Amplification} = \Delta_{\text{compound}} - (\Delta_{\text{hot}} + \Delta_{\text{dry}})$$

  - < 0: compound is worse than the sum ⇒ **synergy**;
  - ≈ 0: **additive**;
  - > 0: less than the sum (**sub-additive**).
- **Uncertainty**: 95% confidence intervals from a *year-block bootstrap*. We resample whole years (1,000 times), because neighbouring cells in the same year are not independent. A result is called significant when its interval excludes zero.

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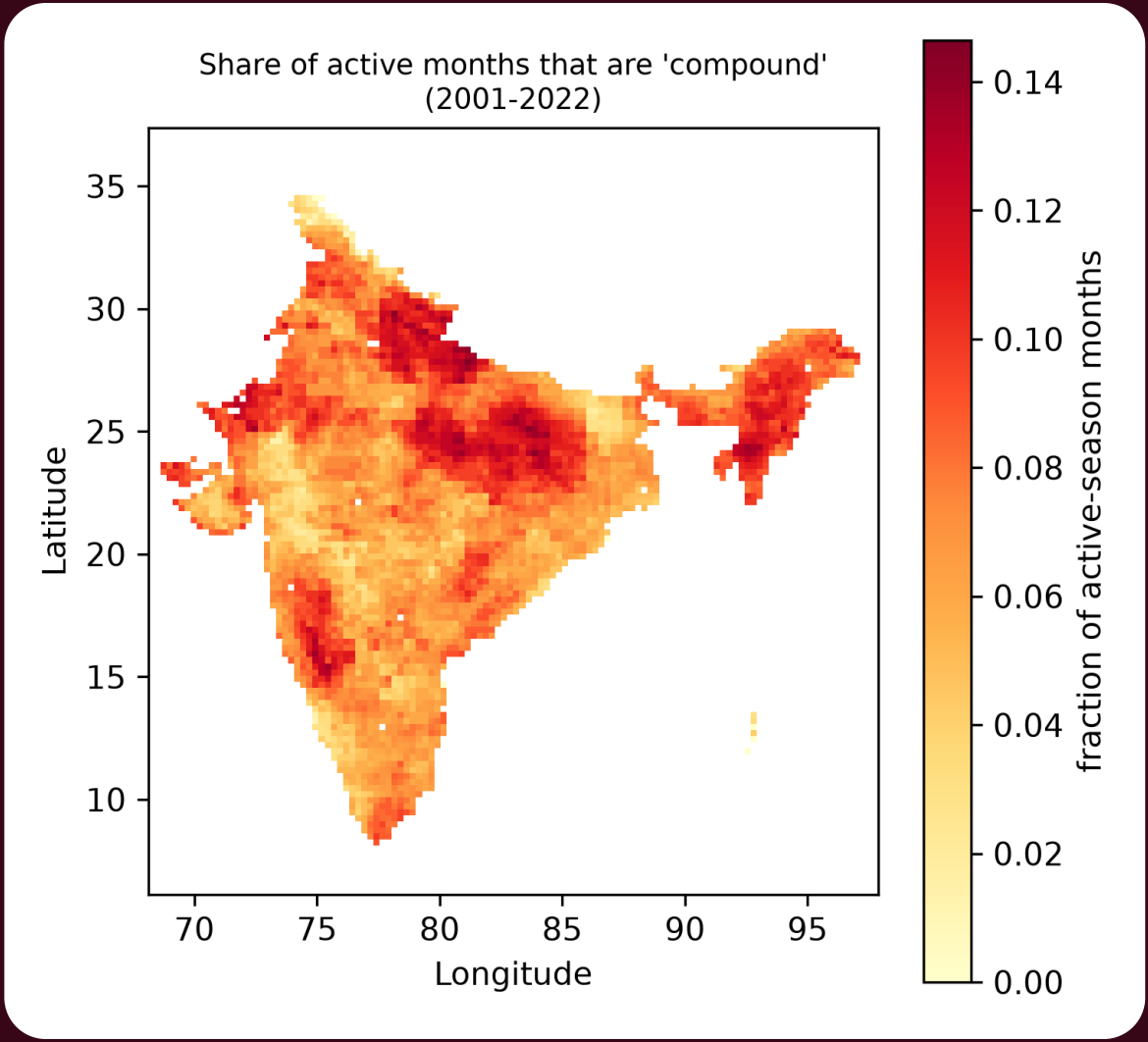
04

PRELIMINARY ANALYSIS (H1, H2)



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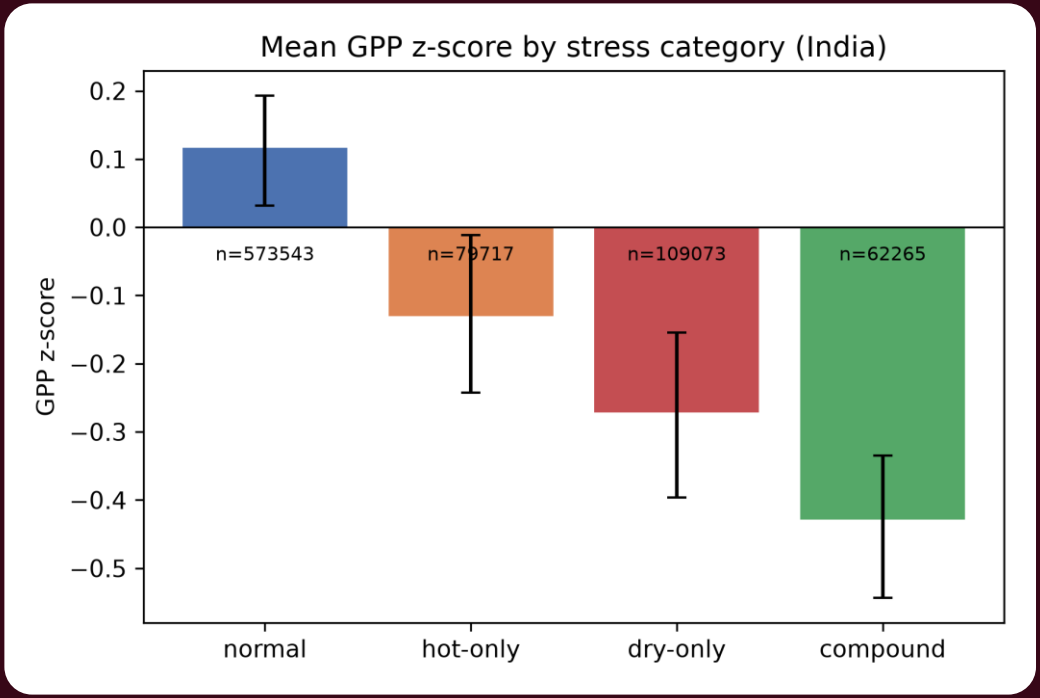
WHERE COMPOUND MONTHS HAPPEN

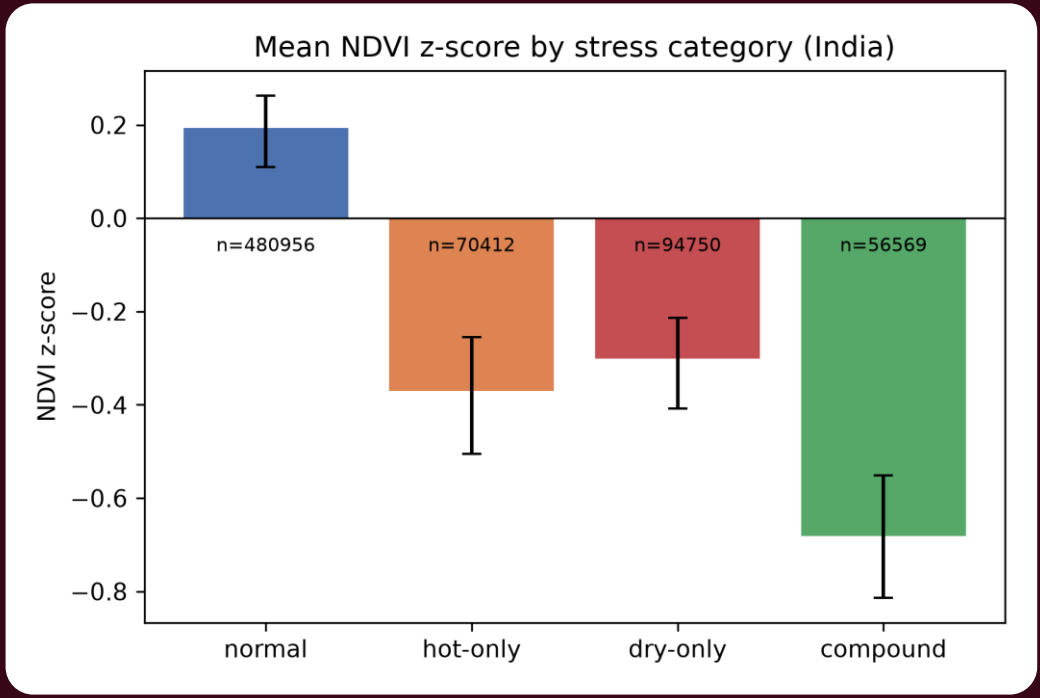


- Share of each cell's growing-season months that were compound, 2001–2022.
- Overall, 7.6% of growing-season grid-months are compound.
- Hotspots: the northern plains, east-central India, the North-East, and a belt in the western Deccan (around 15–18°N).
- Both North and Peninsular India have more than 20,000 compound grid-months, enough for a reliable comparison.

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H1: IS VEGETATION HIT HARDER IN COMPOUND MONTHS?





Mean z [95% CI]	Normal	Hot-only	Dry-only	Compound
GPP	+0.12	−0.13 [−0.24, −0.01]	−0.27 [−0.40, −0.15]	**−0.43** [−0.54, −0.33]
NDVI	+0.19	−0.37 [−0.50, −0.26]	−0.30 [−0.41, −0.21]	**−0.68** [−0.81, −0.55]

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H1: RESULT

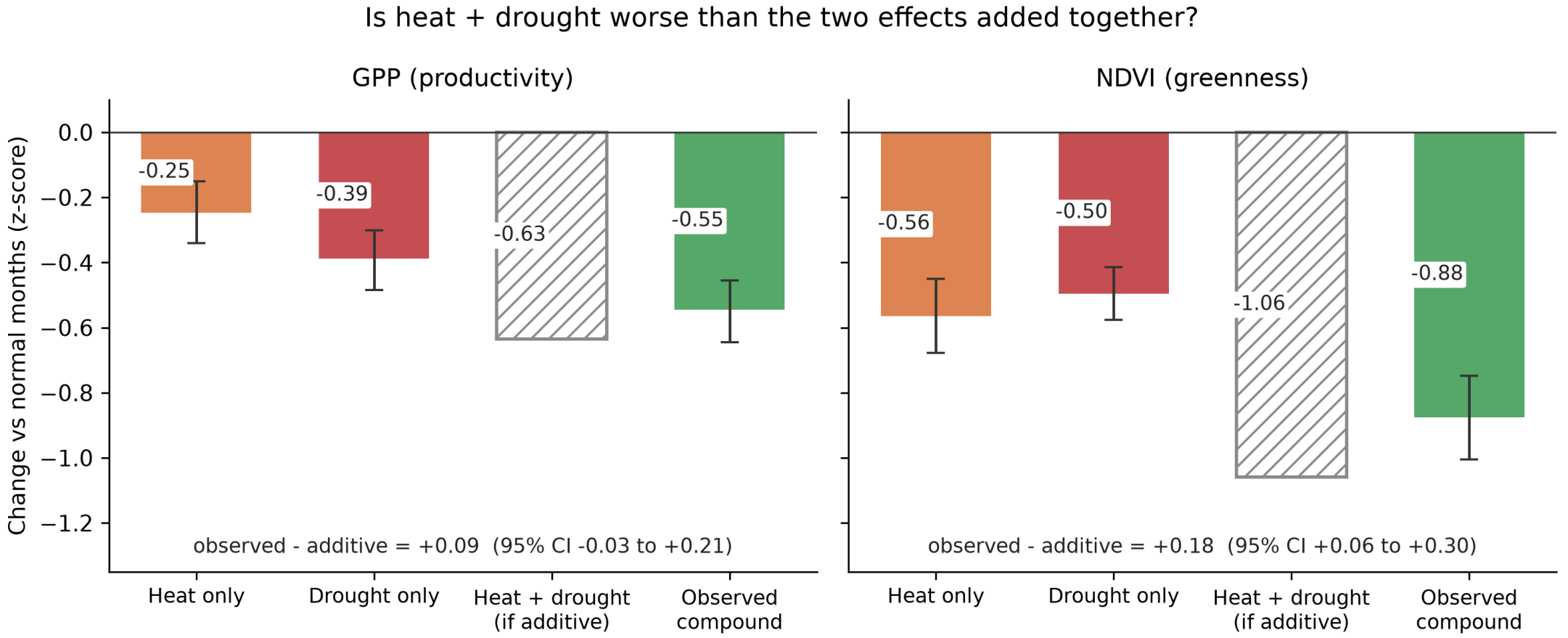
- All three stress types lower productivity and greenness significantly compared with normal months.
- **Compound months are the worst**, and significantly worse than either single stress:
  - compound minus dry-only: GPP −0.16 [−0.26, −0.04], NDVI −0.38 [−0.49, −0.28];
  - compound minus hot-only: GPP −0.30 [−0.40, −0.21], NDVI −0.31 [−0.43, −0.21].
- Robust to the drought index. With rainfall-only SPI-3, compound minus dry-only is GPP −0.26 [−0.37, −0.13] and NDVI −0.48 [−0.61, −0.35].
- **Why?** Compound months combine the driest soils *and* the thirstiest air:

	Normal	Hot-only	Dry-only	Compound
Root-zone soil moisture z	+0.29	−0.75	−0.38	**−1.04**
Air dryness (VPD) z	−0.38	+1.11	+0.18	**+1.43**

H1 SUPPORTED.

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H2: IS THE COMBINATION WORSE THAN THE SUM?



Hatched bar: what we would expect if the two damages simply added up. Green bar: what actually happened in compound months.

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H2: RESULT

Amplification = $\Delta_{compound} - (\Delta_{hot} + \Delta_{dry})$; < 0 would mean synergy.

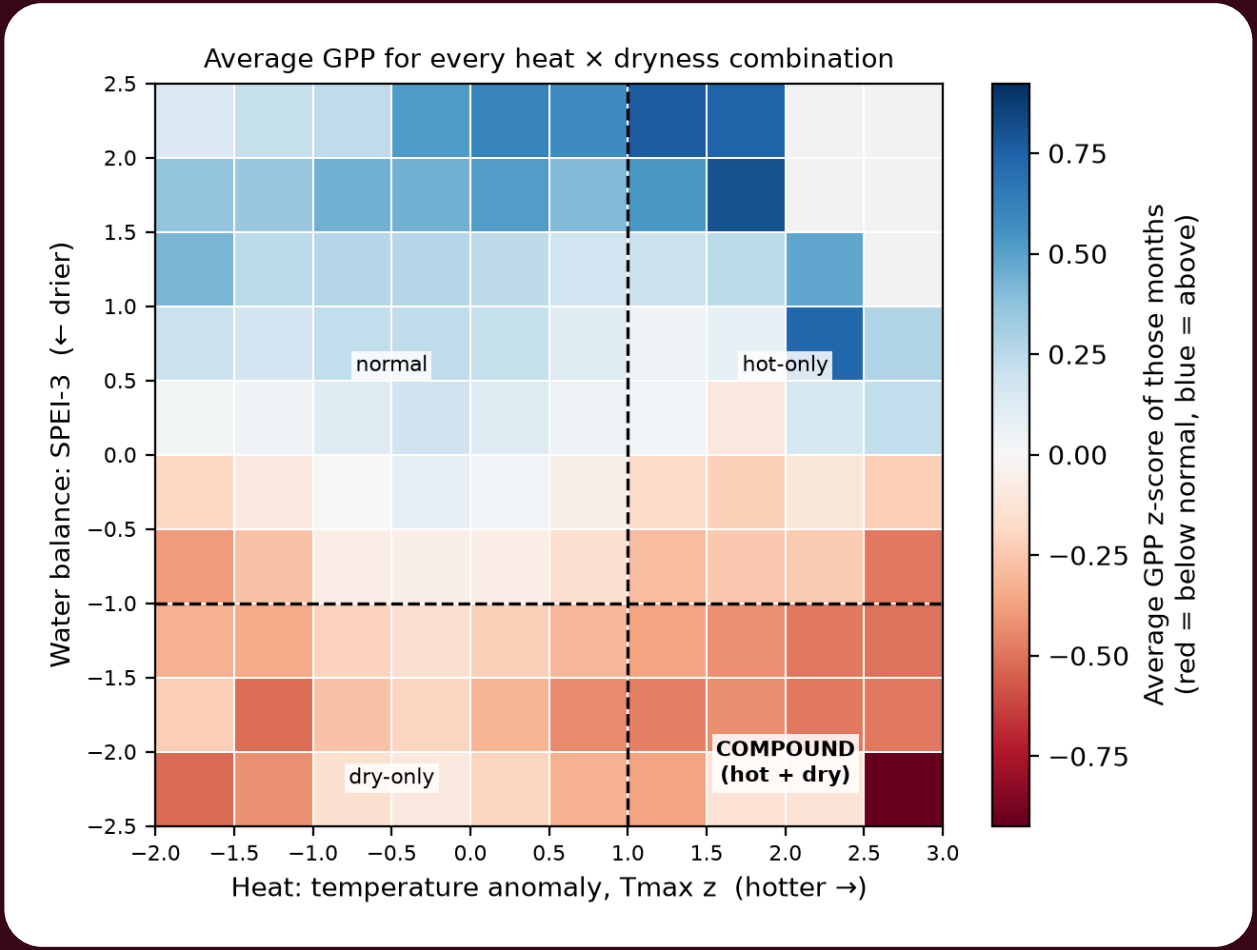
	with SPEI-3	with SPI-3
GPP	+0.09	−0.00
	[−0.03, +0.21]	[−0.12, +0.13]
NDVI	+0.18	+0.06
	[+0.06, +0.30]	[−0.05, +0.18]

- **No synergy**: the amplification is never significantly negative.
- Heat and drought damages roughly **add up**. Compound months are worse than either alone, but not worse than both together.
- With SPEI-3, NDVI is even slightly *less* than additive, but this disappears with SPI-3. **The drought definition matters.**

H2 not supported (for average productivity, India-wide).

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H2: RESPONSE SURFACE (A CHECK WITHOUT BOXES)



WHAT YOU ARE LOOKING AT

- x = heat (Tmax *z*); y = water balance (SPEI-3, lower = drier).
- Each square: average GPP *z* of all months in that range (red = below normal; grey = fewer than 30 months).
- Dashed lines = our cut-offs; bottom-right = the compound box.

WHAT WE SEE

- GPP falls as it gets drier: about +0.5 in the wettest rows, -0.35 in the driest.
- Worst square: hottest + driest corner (-0.92). Hot but wet months are not below normal.
- ⇒ Dryness is the main driver and heat adds to it, consistent with additive effects.

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OBSERVATIONS SO FAR

- **H1 supported.** Compound heat–drought months reduce productivity (GPP) and greenness (NDVI) significantly more than heat alone or drought alone. This holds for both drought indices.
- **H2 not supported.** The compound effect is about the *sum* of the heat and drought effects, not more.
- **Mechanism.** Compound months have the driest soils and the thirstiest air at the same time.
- **Drought definition matters.** SPEI vs SPI changes the H2 details. Hot-only months also have dry soils (soil moisture $z = -0.75$), which rainfall-based indices miss.
- **How this fits the literature.** Meena et al. [4] report "nonlinear amplification" for India using *probabilities* of vegetation drought. Our test on *average* anomalies finds additivity. The two ask different questions; comparing them directly is planned.
- **Caveats.** MOD17 GPP is partly modelled (it uses air dryness and temperature, and has no soil-moisture input). NDVI is from the same satellite. Monthly 0.25° data blur short heatwaves.

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05

THINGS LEFT TO DO



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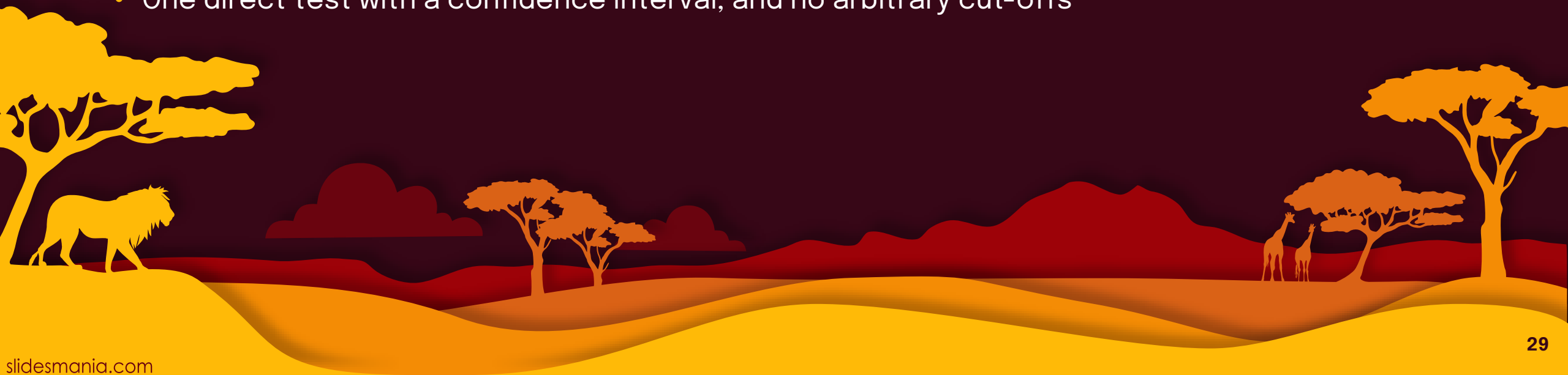
IMPROVING H2: REGRESSION INSTEAD OF FOUR BOXES

WHY AND WHAT

The four boxes use fixed cut-offs (±1) and ignore how strong each anomaly is. So, by the final submission, we will move from four boxes to a **regression fitted to every month's actual heat and dryness values**.

$$\text{GPP} = a \times \text{heat} + b \times \text{dryness} + c \times (\text{heat} \times \text{dryness})$$

- a, b: how much GPP drops per unit of heat or dryness on its own
- $c < 0 \Rightarrow$ heat hurts more when it is also dry (synergy)
- One direct test with a confidence interval, and no arbitrary cut-offs



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FURTHER WORK

H3: RECOVERY

- Time for vegetation to recover after each event
- Compound vs drought-only at the same drought severity

H4: LAND COVER & REGION

- Forests, croplands, grasslands
- Peninsular vs North India

BEYOND INDIA

- Repeat the analysis for another country
- Does India's "additive" result hold elsewhere?

All our datasets cover the globe (CHIRPS: 50°S–50°N), so the same pipeline can be pointed at a new region.

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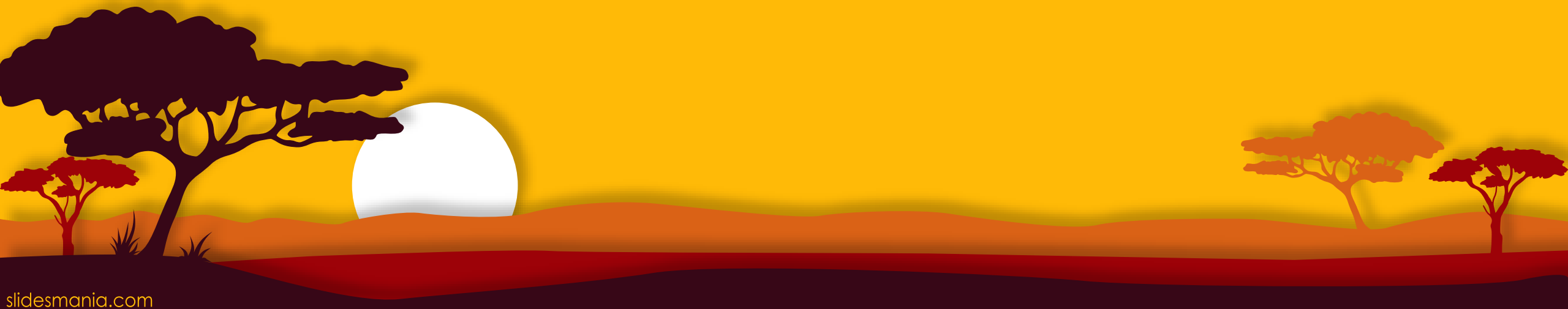
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THANK YOU!

Questions?

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